

Cognitive Radar-Vision Scene Analysis via False Discovery Rate Control

Kognitive Radar-Vision-Szenenanalyse mittels Kontrolle der Falscherkennungsrate

Bachelor thesis in Electrical Engineering and Information Technology

by Lamin Hossain

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Abstract

This is just an example thesis provided by the signal processing group. The abstract should be short and should neither contain abbreviations nor math.

Notation, Symbols and Operators

$f'(\cdot)$	the first derivative of a function $f(\cdot)$
$f_X(x)$	probability density function of X
$F_X(x)$	cumulative distribution function of X
$\mathcal{U}(a, b)$	Uniform distribution between a and b
$\mathcal{N}(\mu, \sigma^2)$	Normal distribution with mean μ and variance σ^2
$\Phi(x)$	cdf of univariate Gaussian distribution
n, N	normal-font letter denotes a scalar
$\mathbf{a}$	bold lowercase letter denotes a vector
$\mathbf{A}$	bold uppercase letter denotes a matrix
$\mathcal{X}$	calligraphic letter denotes a set
$\mathbb{R}$	set of real numbers
$\mathbb{Z}^+$	set of positive integer values
$\mathbb{R}^{r \times 1}$	set of column vectors of size $r \times 1$ on $\mathbb{R}$
$\mathbb{R}^{r \times r}$	set of matrices of size $r \times r$ on $\mathbb{R}$
$\exp(\cdot)$	exponential function
$\ln(\cdot)$	natural logarithm
$\Gamma(\cdot)$	gamma function
$\text{erf}(\cdot)$	error function
$\mathbf{A}^{-1}$	matrix inverse
$\mathbf{A}^\top$	matrix transpose
$ \alpha $	absolute value of scalar
$ \mathbf{A} $	determinate of matrix
$\text{Tr}(\cdot)$	trace of matrix
$\otimes$	Kronecker product

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1 Introduction

The first chapter is usually an introduction. It has no minitoc. Amongst other things it should contain

- a short overview of the treated problem,
- the motivation and relevance,
- an outlook of the chapter arrangement.

2 Fundamentals

2.1 Placeholder 1

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2.2 Signal Reconstruction

As an example, the problem of signal reconstruction is considered:

$$x(n) = s(n) + v(n) \quad n = 0, \dots, N - 1 \quad (2.1)$$

Here, $s(n)$ is any signal of interest and $v(n)$ is a disturbance signal. The goal is to find an estimated version of $s(n)$ based on N noisy measurements $x(n)$. A common approach would be a filtering operation, as depicted in Figure 2.1. The vector $\mathbf{x}$ containing all N measurements is given by

$$\mathbf{x} = [x(0), \dots, x(N - 1)]^\top. \quad (2.2)$$

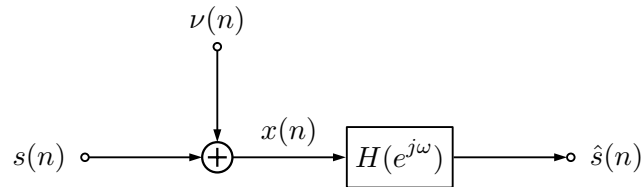


Figure 2.1: Reconstruction of a noisy signal using a filtering operation.

3 State of the Art

3.1 Placeholder 2

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3.2 The Wiener filter

The Wiener filter is an approach which finds an optimal filter $H(e^{j\omega})$ which minimizes $s(n) - \hat{s}(n)$, as shown in Figure 2.1, in the mean square sense. For an uncorrelated $s(n)$ and $\nu(n)$ we have

$$H(e^{j\omega}) = \frac{C_{SS}(e^{j\omega})}{C_{SS}(e^{j\omega}) + C_{VV}(e^{j\omega})} \quad (3.1)$$

where $C_{SS}(e^{j\omega})$ and $C_{VV}(e^{j\omega})$ are the spectra of $s(n)$ and $\nu(n)$, respectively.

4 Methodology

4.1 Placeholder 3

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4.2 The proposed M-Estimator

Our proposed algorithm.

Algorithm 1 M-Estimator

Input: data $\mathbf{x}$, threshold τ
1. Set $j = 1$, **initialize** $\hat{\theta}_0$ **and estimate** $\hat{\sigma}$:
 $\hat{\theta}_0 = \text{median}(\mathbf{x})$
 $\hat{\sigma} = 1.483 \text{ median}(|\mathbf{x} - \text{median}(\mathbf{x})|)$
while $\frac{|\hat{\theta}_j - \hat{\theta}_{j-1}|}{\hat{\sigma}} > \tau$ **do**
 2. Computation of the weights:
 for $i = 1, \dots, L$
 $w_{ji} = W\left(\frac{x_i - \hat{\theta}_j}{\hat{\sigma}}\right)$
 end for
 3. Calculate the estimate $\hat{\theta}_{j+1}$:
 $\hat{\theta}_{j+1} = \frac{\sum_{i=1}^L w_{ji} x_i}{\sum_{i=1}^L w_{ji}}$
 4. Set $j = j + 1$
end while
Output: $\hat{\theta}_j$

5 Results

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6 Discussion

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7 Conclusion and Outlook

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